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Calculating the index of refraction of a plastic hemisphere

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In this paper, we will discuss what an index of refraction is and how to calculate that number. The experiment used a helium-neon gas laser, a plastic hemisphere, and a basic optics ray table by PASCO. Using that equipment, the index of refraction was calculated for the plastic. The refractive index was calculated as $1.7 \pm 2.8 \times 10^{-2}$. This value falls in the range of refractive indices for plastics, which is 1.3 through 1.8.

I. INTRODUCTION

Have you ever taken a pencil or any object, and put it in water and see that the object looks like it has shifted in a weird way in the water? This phenomenon is due to a concept called refraction. Refraction is the bending of light rays as they pass from one medium to another, causing the change of the path of the light rays [1]. For the pencil in water example, the change of medium for light was going from air to water. The water has a different index of refraction than air.

The index of refraction n is a dimensionless number that indicates the light's ability to bend in said medium. The index of refraction also means how fast light travels in that medium [2]. Air has a refractive index of 1.003 and water has a refractive index of 1.333. The higher the refractive index in a material, the more light refracts within.

It may seem trivial to calculate this dimensionless number, but it can be done using the law of refraction. The law of refraction, more commonly known as Snell's law, is used to calculate the index of refraction for different materials, and it is what we used in our experiment to calculate the index of refraction for a plastic hemisphere.

II. DEVELOPMENT

Snell's Law is given by the following equation where n_1 is the index of refraction for the first medium, θ_1 is the incident angle, n_2 is the refractive index for the second medium, and θ_2 is the refracted angle,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2. \tag{1}$$

In our experiment, n_1 will be the refractive index of air, which we will round to and use the value of 1.0.

As the angle of incidence increases, there becomes a critical angle where the angle of refraction $\theta_2 = 90$ deg. When the angle of incidence is greater than the critical angle, refraction is impossible, and the light is completely reflected. The critical angle a_c can be measured visibly in the experiment, or calculated in the following equation,

$$a_c = \arcsin\left(\frac{1}{n}\right), \tag{2}$$

where n is the index of refraction of the material.

III. THE EXPERIMENT

The goal of our experiment was to find the index of refraction for a plastic hemisphere. For our experiment, we used a helium-neon gas laser and directed the beam through the plastic hemisphere. We had the plastic hemisphere placed on a basic optics ray table by PASCO, which made it very convenient to measure the incidence angle and refracted angle.

We took the angle of incidence in increments of 5 degrees from normal from 5 deg. to 35 deg. We decided to not go past 35 deg. because around 40 deg. from normal was the critical angle, and we measured it to be 42 deg. from normal. We repeated this process three times to get an average of the refracted angles.

IV. DATA ANALYSIS

With the data we got from the experiment, and rearranging Snell's law,

$$n_2 = \frac{n_1 \sin\theta_1}{\sin\theta_2}, \tag{3}$$

we can solve for the index of refraction of the plastic. By doing this, we can create a plot with the angle of incidence on the y axis and the refracted angle on the x axis, the index of refraction is 1 over the slope.

The slope of the line can be seen in fig. 1, which is 0.59. Using that we can calculate the index of refraction,

$$\begin{aligned} n_2 &= \frac{1}{0.59}, \\ n_2 &= 1.7 \pm 2.8 \times 10^{-2}. \end{aligned} \tag{4}$$

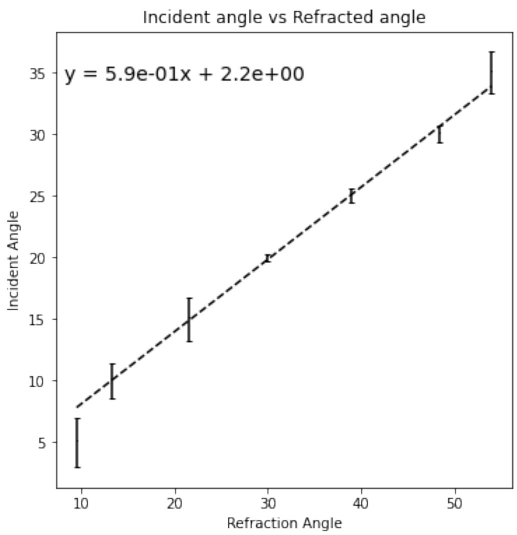

FIG. 1. This is a plot of the incident angle vs refracted angle using the data from our experiment. The index of refraction is 1 over the slope of the line.

Our calculated index of refraction for the plastic we used was $1.7 \pm 2.8 \times 10^{-2}$. The range of refracted indices for most plastics range from 1.3 to 1.8 [3], and our value falls into that range.

Now that we found the index of refraction for our plastic, we can calculate the critical angle using eq. 2.,

$$a_c = 36 \text{ deg.} \tag{5}$$

The value we calculated for the critical is different from the value we measured in the lab, which was 42 deg. This number could be off due to visual errors in the experiment such as recording the wrong number for the refracted angle.

Lastly as stated before, the index of refraction can be used to find the speed light travels in the medium. This can be found using the following equation,

$$v = \frac{c}{n}, \tag{6}$$

where v is the speed of light in the medium, and c is the speed of light. Substituting our value for the refractive index, we can solve for v,

$$v = 1.76 \times 10^8 \text{ m/s}. \tag{7}$$

This means that light travels at about 58% of the speed of light in the plastic

V. CONCLUSION

Using Snell's law we were able to calculate the index of refraction of a plastic hemisphere using a helium-neon gas laser and a basic optics ray table by PASCO. The value we calculated for was $1.7 \pm 2.8 \times 10^{-2}$. Using that we calculated for the critical angle of 36 deg., and the speed which light travels in the plastic, which was 58% of the speed of light.

To get more precise numbers for this experiment, one could use different materials to measure the incident and refracted angle, as that is where the error occurred in our experiment.

To take this experiment further, someone could use different laser colors to see if that yields in different refractive indices. This experiment could also be done to figure out an unknown material. If you can calculate the refractive index of an unknown material, you can compare that number to the values of various other materials and find which number matches the closest.

Appendix A: Uncertainty analysis

1. Data Table

Here is the data table from our experiment. After completing the first trial, we realized we needed to take the angle of incidence in smaller steps, so we made that change for the second and third trials.

2. Uncertainty Calculation

In this experiment, we decided to measure the uncertainty in the angle of refraction. The value we decided to use was ± 1 deg. If we were to do this experiment again, we would find a more precise way to find the angle of refraction, as that is where we found our error.

3. Error Calculation

When calculating the error, we used a least squares model and used the following Python script:

For this script, the angle of refraction was our x value, the angle of incidence was our y value, and dy was the derivative of our refracted angle times the uncertainty. The script would give us our slope, m, intercept, b, error

	Refracted angles			
Incident angle	Trial 1	Trial 2	Trial 3	Average
5		11	8	9.5
10	12	13	15	13.33
15		22	21	21.5
20	29	31	30	30
25		39	39	39
30	47	49	49	48.33
35		58	50	54

TABLE I. Data table

```
def least_squares(x,y,dy):
    w = 1/(dy**2)
    W = sum(1/(dy**2))
    mux = np.average(x,weights=w)
    varx = np.average((x-mux)**2,weights=w)
    muy = np.average(y,weights=w)
    # Least-squares form of m,b
    m = np.average((x-mux)*(y-muy),weights=w)/varx
    b = muy - m*mux
    # Least squares uncertainty
    db = sqrt((1+(mux**2/varx))/W)
    dm = 1/sqrt(varx*W)
    return m,b,dm,db
```

FIG. 2. This is a figure of the least squares error script

in slope, dm, and error in intercept, db. Using those four variables, we were able to calculate the index of refraction as well as the error in that number.

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